

Project - Low Level Design

on

Multi-Agent Automotive System

Course Name: Agentic AI

Medicaps University – Datagami Skill Based Course

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1. Introduction

The Multi-Agent Automotive System is an Agentic AI-based project designed to demonstrate how multiple intelligent agents can collaborate to perform complex tasks efficiently. Instead of relying on a single AI model, the system divides responsibilities among specialised agents to improve accuracy, scalability, and organization of outputs.

In this system, a Researcher Agent is responsible for gathering automotive data such as car specifications, features, and performance details, while a Writer Agent processes this information and presents it in a structured and user-friendly format. These agents communicate and coordinate through defined workflows using tools like Python, CrewAI/LangChain, and LLM APIs.

The project highlights key concepts of Agentic AI, including autonomy, task delegation, inter-agent communication, and workflow orchestration, making it a practical implementation of collaborative AI systems in the automotive domain.

1.1 Scope of the Document

This document presents the Low-Level Design (LLD) of the Multi-Agent Automotive System. It defines the architecture, workflows, components, and communication mechanisms used in the system. The goal is to provide a clear technical blueprint for implementation using Python, LangChain/CrewAI, and LLM APIs.

1.2 Intended Audience

This document is intended for:

- Developers implementing the system
- Project evaluators and mentors
- Students studying Agentic AI systems

1.3 System Overview

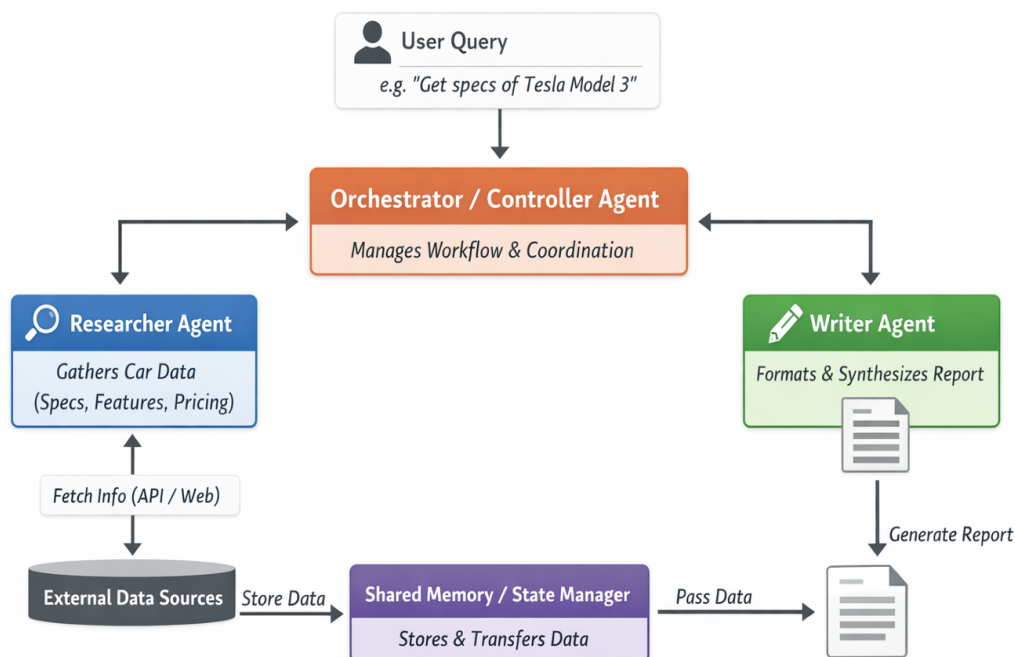
The Multi-Agent Automotive System is designed to simulate a collaborative AI environment where multiple agents work together to accomplish a task.

The system consists of:

- **Researcher Agent:** Responsible for collecting automotive data such as specifications, features, and performance details.
- **Writer Agent:** Responsible for formatting, refining, and presenting the collected data in a structured and readable format.

These agents communicate using defined protocols and share intermediate outputs to complete the task efficiently. The system demonstrates key Agentic AI concepts such as autonomy, task delegation, and inter-agent communication.

Collaborative Automotive Agent Architecture



2. System Design

2.1 Application Design

The system follows a modular multi-agent architecture:

- Input Layer: Accepts user query (e.g., car name)
- Agent Layer: Contains Researcher and Writer agents
- Orchestration Layer: Manages task flow using CrewAI/LangChain
- Output Layer: Displays final structured response

2.2 Process Flow

1. User provides input (car name/query)
2. Task is assigned to Researcher Agent
3. Researcher Agent gathers relevant automotive data
4. Data is passed to Writer Agent
5. Writer Agent formats and refines the content
6. Final output is returned to the user

2.3 Information Flow

- User Input → Researcher Agent
- Researcher Output → Writer Agent
- Writer Output → Final Response

Information is passed as structured text between agents using shared memory or task outputs.

2.4 Components Design

1. Researcher Agent

- Role: Data collection
- Input: Car name
- Output: Raw structured data

2. Writer Agent

- Role: Content formatting
- Input: Raw data from Researcher
- Output: Final formatted report

2.5 Key Design Considerations

- Modularity: Agents are independent and reusable
- Scalability: Additional agents can be added easily
- Maintainability: Clear separation of responsibilities
- Accuracy: Ensuring reliable data generation

2.6 API Catalogue

1. Input API

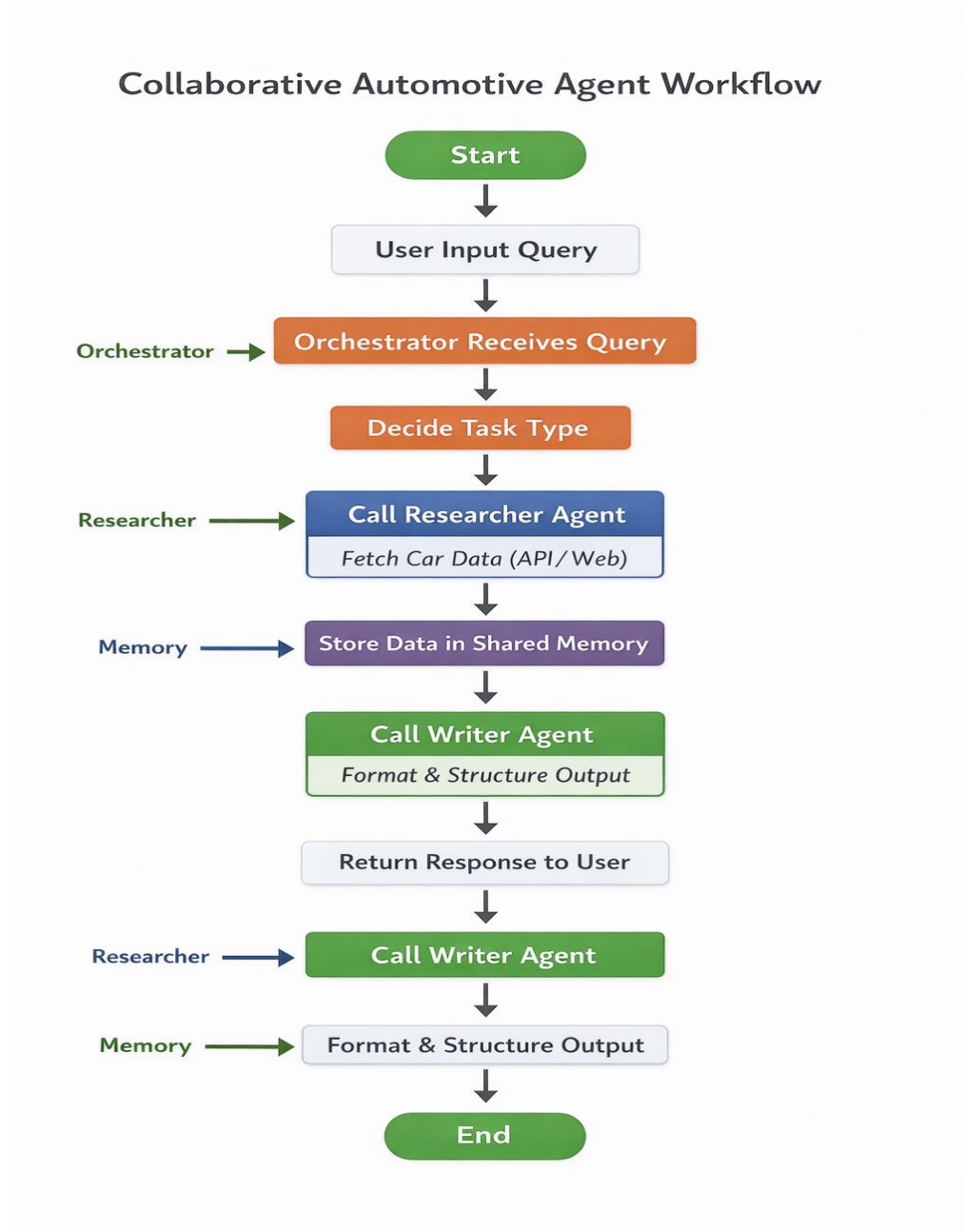
- Function: Accept user query
- Input: String (car name)
- Output: Trigger workflow

2. Research API

- Function: Fetch car data
- Input: Car name
- Output: Raw information

3. Output API

- Function: Display final result
- Input: Formatted content
- Output: User-readable response



3. References

This document is prepared with the help of the following resources:

- LangChain Documentation
- CrewAI Documentation
- Python Official Documentation
- OpenAI API Documentation

These references helped in understanding and designing the system.